

Multi-Modal EEG Feature Fusion for Upper-Limb Motor State Classification: A Comparative Deep Learning Study

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Abstract

Decoding complex upper-limb motor intent (Rest, Elbow, Hand, Forearm) from non-invasive scalp EEG remains a formidable challenge due to volume conduction and severe single-trial noise. In this report, we systematically evaluate a progression of decoding strategies, advancing from classical Common Spatial Patterns (CSP) to deep convolutional baselines (EEGNet). To break a stubborn generalization plateau, we introduce multi-modal dual-branch architectures fusing raw temporal epochs with frequency-domain features, including Welch Power Spectral Density (PSD) and wavelet-based Cross-Spectral Density (CSD) [1, 2]. Crucially, our quantitative analysis uncovers a “dual-branch dilemma”: high-capacity temporal convolutions aggressively overfit to single-trial noise, actively suppressing the highly discriminative power of the connectivity features. Ultimately, we demonstrate that projecting functional connectivity matrices onto a Riemannian Tangent Space and extracting topological Node Degrees overcomes this bottleneck, offering a highly-regularized, geometrically sound framework for robust motor decoding.

Index Terms—EEG motor decoding; common spatial patterns; deep learning; multimodal fusion; upper-limb classification.

1 Introduction

Decoding multiple upper-limb movement states from scalp EEG motivates brain–computer interfaces, neurorehabilitation, and robotic assistive devices [2]. Signal quality at the scalp is limited by spatial mixing (volume conduction), low SNR, and variability between trials, which complicates multiclass decoding.

Purpose of this work. We organise the approach as two blocks applied to identical cue-locked data. **Block I** is a classical *CSP+LDA pipeline*: handcrafted spatial filters optimised for discriminative variance, followed by a linear Gaussian classifier [1].

Block II is a set of *convolutional decoders*: an EEGNet-style architecture [3] operating on raw voltage epochs together with two custom dual-branch models that concatenate or softly gate auxiliary tensors derived from Welch spectral maps and pairwise wavelet cross-spectral features.

Contribution. The comparison isolates where accuracy gains originate when moving from linear spatial prototypes to differentiable temporal receptive fields and, subsequently, multimodal fusion. Section 2 lays out methodology in that order (CSP, then convolutional architectures); Section 3 reports withheld accuracy and accompanying figures; Section 4 interprets them.

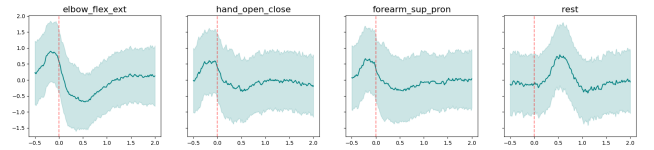


Figure 1: Grand-average cue-locked EEG for four behavioural classes.

Data. Experiments reuse the Graz movement–execution corpus [2]. Code and preprocessing defaults are documented in a github repository referenced in the Acknowledgment section.

Figure 1 shows condition-averaged sensorimotor waveforms obtained after filtering and segmentation.

2 Materials and methods

2.1 Dataset and common preprocessing

We analysed openly released movement–execution EEG [2] recorded while volunteers sustained upper-limb actions with elbow, forearm, hand synergies, and resting fixation (BNCI Horizon 2020, accession 001-2017; 61 electrodes at 512 Hz [2]).

Experiments retained twenty-one peri-central electrodes, resampled the continuous signal at 128 Hz, and extracted cues locked to $[-0.5, 2.0]$ s relative to the stimulus. Trials with peak EEG

magnitude outside $\pm 200 \mu\text{V}$ prior to augmentation were discarded. Mirror movement directions (flexion/extension, supination/pronation, open/close) were pooled, yielding four classes: *elbow*, *hand*, *forearm*, and *rest*.

All trainable architectures shared one stratified random partition of pooled subjects: $\sim 85\%$ of trials optimisation, $\sim 15\%$ validation, fixed pseudo-random seed for comparable convergence noise.

Auxiliary spectral and connectivity tensors. After per-trial, per-electrode temporal normalization, Welch spectra between 3 and 30 Hz constitute log-magnitude PSD maps. Connectivity descriptors arise from modulus of pairwise complex Morlet cross-spectral densities at 10 and 22 Hz summed over unordered electrode pairs (then globally standardised). These tensors feed only the multimodal convolutional hybrids.

Broadband filtering. Recording segments were mains-notch filtered followed by FIR band-pass filtering between 0.5 and 40 Hz (zero-phase MNE kernels). CNN exports omitted common-average re-referencing whereas supplementary CSP optimisation sweeps exercised alternative motor-band hypotheses; divergence between those variants is flagged in Discussion.

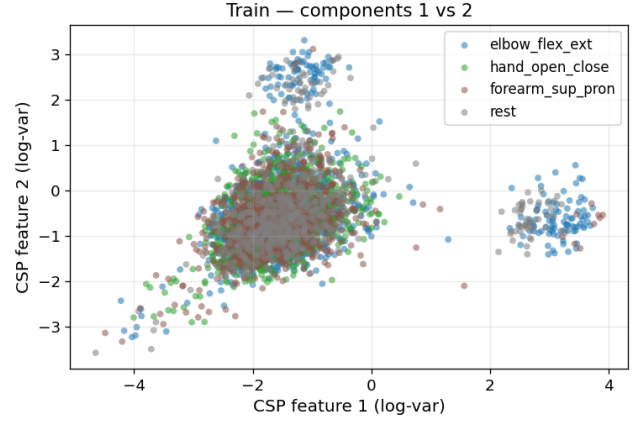
2.2 Block I — CSP with LDA

Block I implements the canonical motor-BCI blueprint: CSP spatial filters optimised for logarithmic-band-power variance ratios and immediately followed by linear discriminant analysis [1]. Six orthogonal spatial patterns provided feature vectors $\mathbb{R}^6$ for LDA.

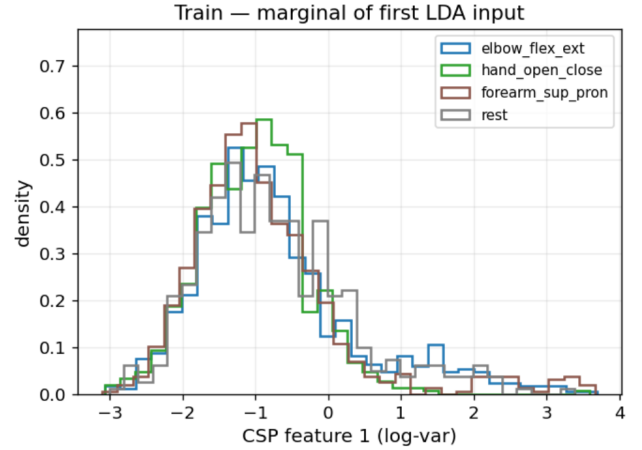
2.3 Block II — convolutional EEG networks

Block II instantiates differentiable temporal receptive fields analogous to EEGNet [3]:

1. **Baseline CNN** — separable convolutional stem processing raw EEG tensors [3].
2. **Spectral concat fusion** (EEGPSNet style) — late fusion by vector concatenation of CNN embeddings with shallow spectral CNN embeddings on Welch maps, batch-normalised prior to softmax.
3. **Connectivity attention fusion** (EEGConnectivityNet style) — independent encoders squeeze waveform embeddings and coherence tensors, softmax attention mixtures modulate



(a) CSP scattergram of logarithmic spectral energies (two projection axes illustrative).



(b) CSP marginal densities for exemplar discriminative component versus class histogram.

Figure 2: Ancillary exploratory plots for Block I feature geometry.

modality weights, lastly MLP softmax classification.

All deep architectures were trained by minimising weighted cross-entropy with mild label smoothing, using adaptive moment (AdamW-type) updates with weight decay, reducing learning rates on withheld-loss plateaus, applying additive Gaussian EEG noise augmentation, and clipping gradients. Checkpoints retaining maximum withheld-set accuracy were kept for inference.

2.4 Ancillary visualisation protocols

Withheld softmax precursors underwent two-dimensional t-SNE with matched hyperparameters across Block II models. Connectivity scalp maps summarise trial-level coupling-derived quantities averaged separately for pooled hand epochs and pooled rest epochs prior to planar interpolation [2].

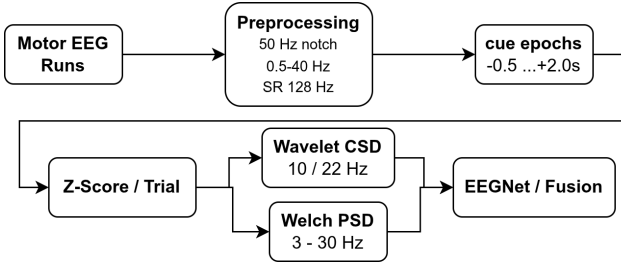


Figure 3: Simplified schematic of preprocessing branches feeding Block II deep models (PSD and connectivity paths optional depending on variant).

3 Experimental results

3.1 Stratification and accuracy benchmarks

Models were judged on withheld validation accuracy from the stratified pooled-subject split described in Section 2.1. CNN checkpoints correspond to maximal validation accuracy during optimisation. CSP+LDA occupies the pooled training partition for estimation and withheld validation for unbiased accuracy.

3.2 Connectivity scalp contrast maps

Figure 4 summarizes trial-averaged connectivity-derived differences between pooled hand synergy epochs and rest on the standard Graz electrode layout [2]. Placement uses a single-column raster width with a tight “here” float anchor so it typically appears immediately beneath this subsection rather than drifting to later pages.

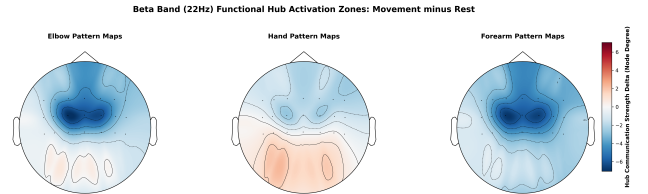


Figure 4: Trial-averaged connectivity-derived spatial patterns contrasting hand synergy epochs versus rest (Graz layout [2]).

3.3 t -Distributed stochastic neighbour embedding of softmax precursors

Figure 5 occupies the full text width (two-column span) so the three Block II embedding panels remain legible; star-floats normally appear at the *top* or *bottom* of a page when sufficient space opens. Embeddings apply identical t -SNE hyperparameters to withheld-trial softmax input vectors across architectures.

Geometric overlaps and linkage to confusion structure are deferred to Section 4.

4 Discussion

Block I versus Block II. Table 1 reports narrow-band exploratory CSP near ~ 38 – 39% withheld accuracy, broadband CSP on tensors sharing the ~ 0.5 – 40 Hz envelope with convolutional preprocessing at $\sim 32\%$, and Block II models on identical splits [1,3]. Convolutional receptive fields improve by ~ 17 – 18 percentage points versus the optimised shallow CSP line and ~ 24 versus broadband CSP paired to identical envelopes. This pattern sug-

Table 1: Held-out four-class classification accuracy on the withheld validation fraction.

Approach series	Representations & notes	Accuracy [%]
Block I CSP+LDA [1]	Motor-band CSP after search across band hypotheses.	~ 36
Block II CNN baseline [3]	Raw EEG via separable temporal convolutions (EEGNet-class baseline).	~ 56
Block II CNN + PSD (EEGSPD- Net)	Late fusion: raw epochs and Welch PSD maps.	~ 61
Block II CNN + CSD attn. (ConnNet)	Raw epochs and coherence tensors with modality attention.	~ 61 – 62

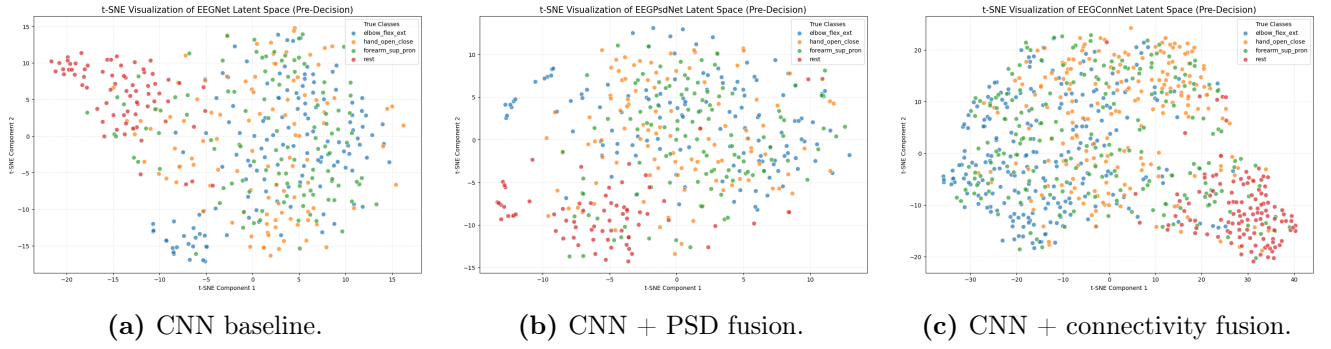


Figure 5: *t*-SNE embeddings of withheld-trial softmax input vectors.

gests discriminative cues that exceed short CSP log-variance summaries of surface waveforms [1].

CSP preprocessing parity. The CSP rows entail different contracts: exploratory fits concentrate variance inside motor-frequency hypotheses, whereas broadband CSP aligns band limiting with Block II. Separately, tensor preparation here omitted common-average referencing for convolutional pipelines [1]. Fair CSP–CNN benchmarking must therefore unify band-pass envelopes, referencing, and cropping before blaming spatial filtering deficits alone.

Multimodal fusion. Welch concatenation and coherence attention each add roughly five percentage points over the EEGNet-style baseline [3]. Auxiliary maps likely supply marginal complementary constraints rather than orthogonal geometry.

Neurophysiological structure versus accuracy. Figure 4 exhibits organised peri-central contrast between trial-averaged hand-related epochs and rest [2] while Table 1 shows only modest incremental gains from fusion. Interpretable connectivity structure and softmax margins are therefore not interchangeable metrics.

Embeddings and residual errors. Colour intermixing in Fig. 5 matches the numeric picture in Table 1: withheld softmax inputs from different classes remain neighbour points for every Block II model, leaving systematic confusions among kinematically related synergies.

Generalisation scope. Withholding trials from a pooled subject mix is optimistic relative to leave-one-subject-out evaluation recommended for neurotechnology translation [2].

5 Conclusion

We contrasted Block I CSP+LDA [1] with Block II convolutional decoders [3] using identical pooled-stratified withheld partitions [2].

Table 1 summarises withheld accuracies; interpretation of embeddings, topographies, fusion limits, and generalisation occupies Section 4.

Recommended follow-ups are aligned preprocessing for classical and deep pipelines, strictly subject-independent benchmarks [2], and fusion designs that exploit auxiliary tensors without weakening temporal discriminability.

Acknowledgment

Code:

<https://github.com/MNR-MarcosOriolPagonabarraga-projects/HMI-EEGMovementClassification.git>

References

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